

KONOTOP, Ukraine

WMO: 332610

Lat: 51.2408N

Lon: 33.1869E

Elev: 149

StdP: 99.55

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-16.7	-22.8	0.5	-19.1	-20.1	0.6	-16.3	7.2	-6.8	6.4	-5.7	1.5	50	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	31.2	20.4	29.5	19.7	27.8	18.9	21.7	28.5	20.8	27.3	20.0	25.9	2.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.5	24.7	18.6	13.7	23.6	17.8	13.0	22.9	63.9	28.7	60.7	27.4	57.9	26.1	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.5	-22.5	33.8	3.3	2.0	-24.9	35.2	-26.8	36.4	-28.7	37.5	-31.1	39.0		
(5)			-22.8	23.1	3.3	1.2	-25.1	23.9	-27.1	24.6	-28.9	25.2	-31.3	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-5.0	-3.9	1.0	9.3	15.4	18.8	20.8	19.6	13.9	7.5	2.0	-3.0		
	DBStd	10.32	5.74	5.92	4.61	4.42	4.58	3.62	3.09	3.54	4.07	4.37	4.69	5.62		
	HDD10.0	1969	466	391	281	66	8	0	0	0	11	102	243	403		
	HDD18.3	3980	724	624	538	273	114	39	12	27	141	336	491	662		
	CDD10.0	1269	0	0	1	43	176	265	333	297	129	24	2	0		
	CDD18.3	238	0	0	0	1	23	53	87	65	9	0	0	0		
	CDH23.3	2111	0	0	0	9	216	432	718	639	97	1	0	0		
	CDH26.7	608	0	0	0	1	47	103	217	221	18	0	0	0		
(14) Wind	WSAvg	2.1	2.4	2.4	2.6	2.3	2.0	1.9	1.7	1.6	1.7	1.9	2.3	2.4		
(15) Precipitation	PrecAvg	601	46	39	40	40	57	63	72	54	53	47	45	44		
	PrecMax	866	80	102	121	93	135	143	198	152	144	105	97	96		
	PrecMin	465	12	8	1	3	20	19	19	2	2	3	7	8		
	PrecStd	97	17	19	26	24	27	28	40	35	35	28	23	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.6	15.5	24.5	30.3	31.4	33.6	34.7	28.9	22.2	14.5	8.9		
		MCWB	5.0	5.4	8.3	13.8	18.2	19.7	21.2	20.3	17.7	15.1	12.3	7.9		
	2%	DB	4.1	5.0	12.0	21.5	27.9	29.4	31.0	31.5	26.2	18.9	11.4	6.7		
		MCWB	3.0	3.8	6.9	12.6	17.2	19.7	20.7	20.7	16.7	13.9	9.7	5.9		
	5%	DB	2.7	3.5	9.4	19.0	25.8	27.7	29.2	29.0	23.5	15.9	9.6	4.7		
		MCWB	1.8	2.4	5.7	11.6	16.6	18.9	20.2	19.5	15.8	12.6	8.4	4.0		
	10%	DB	1.5	2.4	7.1	16.9	23.6	26.0	27.3	26.8	21.1	13.7	8.1	3.2		
		MCWB	0.8	1.5	4.3	10.5	15.7	18.3	19.6	18.4	14.8	10.9	7.1	2.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	5.8	9.6	15.1	20.0	21.9	22.9	22.6	19.3	16.5	12.3	8.0		
		MCDB	5.9	7.1	14.4	21.6	27.0	27.5	29.7	29.8	26.0	20.4	14.3	8.8		
	2%	WB	3.2	3.8	7.5	13.4	18.6	20.7	21.9	21.5	17.6	14.4	10.1	5.9		
		MCDB	3.8	4.7	10.7	19.0	25.3	27.1	28.8	29.0	24.1	17.7	11.1	6.5		
	5%	WB	1.9	2.6	6.0	12.4	17.5	19.8	21.1	20.4	16.5	12.8	8.5	4.1		
		MCDB	2.6	3.3	9.1	17.5	23.7	26.0	27.4	27.4	21.7	15.6	9.4	4.6		
	10%	WB	0.9	1.7	4.6	11.4	16.5	19.0	20.3	19.4	15.4	11.4	7.2	2.6		
		MCDB	1.5	2.3	7.0	16.1	22.2	24.6	25.8	25.3	19.9	13.3	8.0	3.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.2	6.6	9.1	10.4	9.6	9.7	10.8	9.5	7.5	4.4	4.0		
		MCDBR	3.4	4.8	10.1	12.3	12.9	11.9	12.2	14.1	13.5	11.3	6.0	4.2		
	5% WB	MCWBR	3.1	4.0	6.2	5.8	5.0	4.7	4.4	5.0	5.9	6.5	4.7	3.8		
		MCWBR	3.5	4.6	9.0	10.5	11.3	10.5	10.7	12.4	11.6	9.8	5.2	3.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.306	0.338	0.369	0.440	0.406	0.393	0.427	0.429	0.399	0.361	0.341	0.311		
		taud	2.159	2.089	2.150	2.094	2.273	2.331	2.254	2.231	2.275	2.350	2.276	2.213		
	Ebn at Noon	Ebn at Noon	680	758	815	795	848	863	824	799	779	739	641	612		
		Edh at Noon	75	106	121	144	126	120	128	125	107	82	68	62		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.86	1.73	3.03	4.25	5.69	6.03	5.76	5.06	3.36	1.83	0.75	0.54		
	RadStd	RadStd	0.17	0.27	0.49	0.38	0.49	0.51	0.36	0.40	0.42	0.29	0.14	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.75	N/A	N/A	+0.93	+0.52	N/A	-152	-211	+116	+50
(47) Regional (7 neighbors)		+0.70	N/A	N/A	+0.98	N/A	N/A	-138	-198	+114	+55

Nomenclature: See separate page